Lecture 27: Kalman Filter (Optional)

1 Announcement

The material covered in these lecture notes will not be included in the final exam.

2 State Space Models

State space model provides a flexible approach in time series analysis. They are used when the quantity of interest is not perfectly observed, but we only get access to a noisy version of it. Consider the univariate time series y_t satisfying

$$\begin{aligned} y_t &= \mu_t + e_t \\ \mu_{t+1} &= \mu_t + \eta_t, \end{aligned} \tag{1}$$

where $\{e_t\}$ and $\{\eta_t\}$ are two independent Gaussian white noise series and $t = 1, \ldots, T$. The initial value μ_1 may be given or follows a known distribution, and is independent of $\{e_t\}$ and $\{\eta_t\}$ for $t > 0$. In the equation above, μ_t is a pure random walk with initial value μ_1, and y_t is an observed version of μ_t with added noise e_t. In the literature μ_t is referred to as the trend of the series, which is *not* directly observable, and y_t is the *observed* data with observational noise e_t. Notice that the dynamic dependence of y_t is governed by that of μ_t, because $\{e_t\}$ is not serially correlated.

The model specified in Eq. (1) can be used to analyze the realized volatility of an asset price (see Example 11.1 in Tsay's book for details). We think of μ_t as the *true* log-volatility of the asset price, and y_t as the logarithm of realized volatility. The true log-volatility is not directly observed, but evolves over time following a random-walk model. The observed volatility y_t instead is constructed from high-frequency transactions data. For instance, the daily realized volatility used can be obtained using the sum of squares of intraday 10-minute log returns measured in percentage (typically, no overnight returns or the first 10-minute intra-day returns are used, and we refer to Chapter 3 of Tsay's book for more details about realized volatility). This construction is subject to market microstructure noises. Such noise captures a variety of frictions inherent in the trading process: bid–ask bounces, discreteness of price changes, differences in trade sizes or informational content of price changes, gradual response of prices to a block trade, inventory control effects, etc... The standard deviation of e_t denotes the scale used to measure the impact of market microstructure noises.

The model in Eq. (1) is a linear Gaussian state-space model. The variable μ_t is called the state of the system at time t and is not directly observed. The equation for y_t provides the link between the data y_t and the (hidden) state μ_t and is called the *observation equation* with measurement error e_t. The equation for μ_t governs the time evolution of the state variable and is the *state equation* (or *state transition equation*) with innovation η_t.

2.1 Statistical Inference

The objective of the analysis is to infer properties of the state μ_t from the data $\{y_t\ t = 1, \ldots, T\}$ and the model. There are three types of inferences that are commonly discussed in the literature: *filtering, prediction*, and *smoothing*. Let $\mathcal{F}_t = \{y_1, \ldots, y_t\}$ be the information available at time t (inclusive), and assume that the model is known, including all parameters. Briefly, the three types of inferences can be described as follows:

- *Filtering*. Filtering means to recover the state variable μ_t given $\mathcal{F}_t$, that is to remove the measurement errors from data.
- *Prediction*. Prediction means to forecast μ_{t+h} or y_{t+h} for $h > 0$ given $\mathcal{F}_t$, where t is the forecast origin.
- *Smoothing*. Smoothing is to estimate μ_t given $\mathcal{F}_T$, where $T > t$.

For a more precise description of the inference, let us first introduce some notation. Let $\mu_{t|j} = \mathbb{E}[\mu_t|\mathcal{F}_j]$ and $\Sigma_{t|j} = Var(\mu_t|\mathcal{F}_j)$ be, respectively, the conditional mean and conditional variance of the actual (hidden) state μ_t given $\mathcal{F}_j$. Similarly, let us use $y_{t|j}$ to denote the conditional mean of y_t given $\mathcal{F}_j$. Moreover, let $v_t = y_t - y_{t|t-1}$ be the 1-step ahead forecast error for the observation, and $V_t = Var(v_t|\mathcal{F}_{t-1})$ the corresponding conditional variance. From Eq. (1), we obtain that

$$y_{t|t-1} := \mathbb{E}[y_t|\mathcal{F}_{t-1}] = \mathbb{E}[\mu_t + e_t|\mathcal{F}_{t-1}] = \mathbb{E}[\mu_t|\mathcal{F}_{t-1}] = \mu_{t|t-1},$$

where above we have used that $\{e_t\}$ is a white noise series. Consequently, we obtain that

$$v_t = y_t - y_{t|t-1} = y_t - \mu_{t|t-1},$$

and

$$\begin{aligned} V_t &= Var(y_t - \mu_{t|t-1}|\mathcal{F}_{t-1}) = Var(\mu_t + e_t - \mu_{t|t-1}|\mathcal{F}_{t-1}) \\ &= Var(\mu_t - \mu_{t|t-1}|\mathcal{F}_{t-1}) + Var(e_t|\mathcal{F}_{t-1}) = \Sigma_{t|t-1} + \sigma_e^2, \end{aligned} \tag{2}$$

where we have used that, conditional on $\mathcal{F}_{t-1}$, e_t is independent of $\mu_t - \mu_{t|t-1}$. It can also be easily seen that

$$\begin{aligned} \mathbb{E}[v_t] &= \mathbb{E}\left[\mathbb{E}\left[v_t|\mathcal{F}_{t-1}\right]\right] = \mathbb{E}\left[\mathbb{E}\left[y_t - y_{t|t-1}|\mathcal{F}_{t-1}\right]\right] = \mathbb{E}\left[y_{t|t-1} - y_{t|t-1}\right] = 0 \\ Cov(v_t, y_j) &= \mathbb{E}[v_t y_j] = \mathbb{E}\left[\mathbb{E}\left[v_t y_j|\mathcal{F}_{t-1}\right]\right] = \mathbb{E}\left[y_j \mathbb{E}\left[v_t|\mathcal{F}_{t-1}\right]\right] = 0, \qquad j < t. \end{aligned}$$

Hence, the 1-step ahead forecast error v_t is uncorrelated (and hence independent given that the system is Gaussian) with y_j, for $j < t$.

We next state, without proof, a theorem in multivariate statistics that will be useful to derive the Kalman filter equations

Theorem 2.1. *Suppose that $\mathbf{x}, \mathbf{y}, \mathbf{z}$ are three random vectors such that their joint distribution is multivariate normal. Let μ_x, μ_y, μ_z denote, respectively, the mean of the vector x,y, and z. Assume that the covariance matrices of $\mathbf{x}, \mathbf{y}, \mathbf{z}$ denoted, respectively, by $\mathbf{\Sigma_{xx}}, \mathbf{\Sigma_{yy}}, \mathbf{\Sigma_{zz}}$ are nonsingular, and that $Cov(\mathbf{y}, \mathbf{z}) := \mathbf{\Sigma_{yz}} = 0$. Then,*

- $\mathbb{E}[\mathbf{x}|\mathbf{y}] = \mu_{\mathbf{x}} + \mathbf{\Sigma_{xy}}\mathbf{\Sigma_{yy}^{-1}}(\mathbf{y} - \mu_{\mathbf{y}})$

- $Var\left[\mathbf{x}|\mathbf{y}\right] = \mathbf{\Sigma_{xx}} - \mathbf{\Sigma_{xy}}\mathbf{\Sigma_{yy}^{-1}}\mathbf{\Sigma_{yx}}$
- $\mathbb{E}\left[\mathbf{x}|\mathbf{y},\mathbf{z}\right] = \mathbb{E}\left[\mathbf{x}|\mathbf{y}\right] + \mathbf{\Sigma_{xz}}\mathbf{\Sigma_{zz}^{-1}}(\mathbf{z} - \mu_{\mathbf{z}})$
- $Var\left[\mathbf{x}|\mathbf{y},\mathbf{z}\right] = Var(\mathbf{x}|\mathbf{y}) - \mathbf{\Sigma_{xz}}\mathbf{\Sigma_{zz}^{-1}}\mathbf{\Sigma_{zx}}$.

3 Kalman Filter

The goal of the Kalman filter is to update the knowledge of the hidden state variable μ_t *recursively* once a new observation y_t becomes available. In other words, knowing the conditional distribution of μ_t given $\mathcal{F}_{t-1}$ and the new observation y_t, we want to obtain the conditional distribution of μ_t given $\mathcal{F}_t$. The fundamental insight by Kalman was in exploiting the Gaussian property of the system, to obtaining a formula which uses the predicted mean $\mu_{t|t-1}$ and the new observation y_t to obtain $\mu_{t|t}$, and the predicted variance $\Sigma_{t|t-1}$ to obtain the updated variance $\Sigma_{t|t}$. Hence, these computations are recursively done and *do not require, at any point in time, to revisit the whole history of observations* $\mathcal{F}_t = \{y_1, y_2, \ldots, y_t\}$ to make the updates. Because $\mathcal{F}_t = \{\mathcal{F}_{t-1}, v_t\}$, giving y_t and $\mathcal{F}_{t-1}$ is equivalent to the information set generated by v_t and $\mathcal{F}_{t-1}$. Hence, it suffices to consider the joint conditional distribution of $(\mu_t, v_t)^\top$ given $\mathcal{F}_{t-1}$ before we apply the theorem above.

The conditional distribution of v_t given $\mathcal{F}_{t-1}$ is Gaussian with zero mean and variance given by V_t. Moreover, the distribution of μ_t given $\mathcal{F}_{t-1}$ is Gaussian with mean $\mu_{t|t-1}$ and variance $\Sigma_{t|t-1}$. Next, we need to compute the conditional covariance between μ_t and v_t given $\mathcal{F}_{t-1}$. Using the definition, we obtain

$$\begin{aligned}
Cov(\mu_t, v_t|\mathcal{F}_{t-1}) &= \mathbb{E}\left[\mu_t v_t\middle|\mathcal{F}_{t-1}\right] = \mathbb{E}\left[\mu_t(y_t - \mu_{t|t-1})\middle|\mathcal{F}_{t-1}\right] \\
&= \mathbb{E}\left[\mu_t(\mu_t + e_t - \mu_{t|t-1})\middle|\mathcal{F}_{t-1}\right] \\
&= \mathbb{E}\left[\mu_t(\mu_t - \mu_{t|t-1})\middle|\mathcal{F}_{t-1}\right] + \mathbb{E}\left[\mu_t e_t\middle|\mathcal{F}_{t-1}\right] \\
&= \mathbb{E}\left[\mu_t(\mu_t - \mu_{t|t-1})\middle|\mathcal{F}_{t-1}\right] - \underbrace{\mathbb{E}\left[\mu_{t|t-1}(\mu_t - \mu_{t|t-1})\middle|\mathcal{F}_{t-1}\right]}_{=0} + \mathbb{E}\left[\mu_t e_t\middle|\mathcal{F}_{t-1}\right] \\
&= Var(\mu_t|\mathcal{F}_{t-1}) + \mathbb{E}[e_t]\mu_{t|t-1} \\
&= Var(\mu_t|\mathcal{F}_{t-1})
\end{aligned}$$

We can now apply the theorem conditional on $\mathcal{F}_{t-1}$, by setting $x = \mu_t$, $y = (y_1, y_2, \ldots, y_{t-1})$, and $z = v_t$ in the theorem above, to obtain

$$\begin{aligned}
\mu_{t|t} &= \mu_{t|t-1} + \frac{\Sigma_{t|t-1}}{V_t} v_t := \mu_{t|t-1} + K_t v_t \\
\Sigma_{t|t} &= \Sigma_{t|t-1} - \frac{\Sigma_{t|t-1}^2}{V_t} := \Sigma_{t|t-1}(1 - K_t),
\end{aligned}$$

where $K_t = \frac{\Sigma_{t|t-1}}{V_t}$ is referred to as the Kalman gain and captures the information content of the observation. It quantifies the contribution of the new shock v_t to the estimation of the hidden state variable μ_t. This also corresponds to the regression coefficient of the hidden state μ_t on v_t. The Kalman gain is the fundamental insight of Kalman, and provides a means to update the

filtered estimate of μ_t by correcting the predicted value $\mu_{t|t-1}$ with the information power of the new observation (the information power is quantified by the Kalman gain).

We can then use our filtered estimate of μ_t given $\mathcal{F}_t$ to predict μ_{t+1}. This is achieved via

$$\begin{aligned}\mu_{t+1|t} &= \mathbb{E}\left[\mu_t + \eta_t \middle| \mathcal{F}_t\right] = \mathbb{E}\left[\mu_t \middle| \mathcal{F}_t\right] = \mu_{t|t} \\ \Sigma_{t+1|t} &= Var(\mu_{t+1}\big|\mathcal{F}_t) = Var(\mu_t\big|\mathcal{F}_t) + Var(\eta_t) = \Sigma_{t|t} + \sigma_\eta^2.\end{aligned}$$

Above, we have used the fact that the state noise term η_t is independent of the filtration $\mathcal{F}_t$, which only contains information about the noise terms $\eta_1, \eta_2, \ldots, \eta_{t-1}$. Once the new data y_{t+1} is observed, we can then repeat the procedure above to update our knowledge of μ_{t+1}. Putting altogether, we obtain the Kalman filter equations. Assume that μ_1 is initially distributed as a Gaussian distribution with mean $\mu_{1|0}$ and variance $\Sigma_{1|0}$, where both parameters are given. We then have the following sufficient statistics for the incremental computation of the filter:

$$\begin{aligned}v_t &= y_t - \mu_{t|t-1} \\ V_t &= \Sigma_{t|t-1} + \sigma_e^2 \\ K_t &= \frac{\Sigma_{t|t-1}}{V_t} \\ \mu_{t+1|t} &= \mu_{t|t-1} + K_t v_t \\ \Sigma_{t+1|t} &= \Sigma_{t|t-1}(1 - K_t) + \sigma_\eta^2,\end{aligned}$$

for $t = 1, \ldots, T$.

4 Properties of the Forecast Error and Maximum Likelihood estimation (not included in the final exam)

The properties of the forecast errors v_t are very useful in many applications. Given the initial values $\Sigma_{1|0}$ and $\mu_{1|0}$ which are independent of y_t, the Kalman filter enables to compute v_t recursively as a linear function of $y_1, y_2, \ldots, y_t$. Specifically, we have

$$v_1 = y_1 - \mu_{1|0} \tag{3}$$

$$v_2 = y_2 - \mu_{2|1} = y_2 - \mu_{1|0} - K_1(y_1 - \mu_{1|0}) \tag{4}$$

$$v_3 = y_3 - \mu_{3|2} = y_3 - \mu_{1|0} - K_2(y_2 - \mu_{1|0}) - K_1(1 - K_2)(y_1 - \mu_{1|0}), \tag{5}$$

and so on. We can write this transformation in a matrix form as follows

$$\mathbf{v} = \mathbf{K}(\mathbf{y} - \mu_{\mathbf{1|0}}\mathbf{1_T}), \tag{6}$$

where $\mathbf{v} = (v_1, v_2, \ldots, v_T)'$, $\mathbf{y} = (y_1, y_2, \ldots, y_T)'$, $\mathbf{1}$ is the T-dimensional vector of ones, and $\mathbf{K}$ is a lower triangular matrix

$$\mathbf{K} = \begin{bmatrix} 1 & 0 & 0 & \ldots & 0 \\ k_{21} & 1 & 0 & \ldots & 0 \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ k_{31} & k_{32} & 0 & \ldots & 0 \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ k_{T1} & k_{T2} & k_{T3} & \ldots & 1, \end{bmatrix}$$

where $k_{i,i-1} = -K_{i-1}$ and $k_{ij} = -(1-K_{i-1})(1-K_{i-2})\dots(1-K_{j+1})K_j$ for $i = 2,\dots,T$, and $j = 1,\dots,i-2$. Notice that the Kalman gain does not depend on $\mu_{1|0}$ or the data $\{y_1, y_2, \dots, y_t\}$. It depends on $\Sigma_{1|0}, \sigma_e^2$ and σ_η^2. This transformation facilitates the maximum likelihood estimation procedure. Importantly, $v_1, v_2, \dots, v_t$ are mutually independent under the Gaussian assumption. To see this, consider the joint probability density function of the data

$$p(y_1, y_2, \dots, y_T) = p(y_1) \prod_{j=2}^{T} p(y_j | \mathcal{F}_{j-1})$$

Eq. (6) indicates that the determination of the Jacobian $\mathbf{K}$ of the transformation is one (the determinant of a lower triangular matrix is the product of its diagonal entries). Moreover, $\mu_{1|0}$ is given, thus $p(v_1) = p(y_1)$. Hence, the joint probability density function of $\mathbf{v}$ is given by

$$p(\mathbf{v}) = p(y_1)\Pi_{j=2}^{T} p(y_j | \mathcal{F}_{j-1}) = p(v_1) \prod_{j=2}^{T} p(v_j | \mathcal{F}_{j-1}) = \prod_{j=1}^{T} p(v_j),$$

where we have used the property shown in last lecture that $Cov(v_t, y_j) = 0$ for all $j < t$. This independent property simplifies the maximum likelihood estimation procedure of the variance of the noise terms. First,

$$p(y_1, y_2, \dots, y_T | \sigma_e, \sigma_\eta) = p(y_1) \prod_{t=2}^{T} p(v_t | \mathcal{F}_{t-1}, \sigma_e, \sigma_\eta),$$

where y_1 is Gaussian with mean $\mu_{1|0}$ and variance V_1, and v_t is Gaussian with mean 0 and variance V_t. Assuming $\mu_{1|0}$ and $\Sigma_{1|0}$ to be known and taking logarithms, we have

$$\log L(\sigma_e, \sigma_\eta) = -\frac{T}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{T} \log(V_t) + \frac{v_t^2}{V_t}.$$

The above likelihood function can be evaluated recursively via the Kalman filter. Many software packages perform state space estimation via a Kalman filter algorithm.